Ukrainian Catholic University

AI Course · Semester Project Proposal · 2026

Text-Guided Target Tracking with a Pan-Tilt Camera under Slow Open-Vocabulary Detection

A latency-aware tracking pipeline for a UGV camera gimbal: asynchronous open-vocabulary detector, fast tracker, Kalman prediction and CLIP-based re-acquisition

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1. Problem and motivation

Open-vocabulary detectors take a text prompt (for example, “a red backpack”) and localize matching objects without task-specific training [1, 2]. They are attractive for a UGV because the target can be changed by typing a sentence instead of collecting data and retraining. However, they are a poor fit for a camera that has to *follow* a target. On edge hardware such detectors are expected to run at only a few frames per second and with a noticeable delay, while a Pan-Tilt gimbal needs a smooth control signal at camera rate. Driving the gimbal directly from the latest detection makes it lag behind a moving target, oscillate, or lose the target completely when it is briefly occluded or leaves the frame. In addition, a detector only knows the *class* described by the text. If several similar objects are in the scene, it does not know which one is the target.

The scope is deliberately narrow: the chassis is stationary or driven by a remote operator, and autonomous navigation is out of scope. The research question is: *how well can a slow, text-conditioned detector keep a Pan-Tilt camera locked on one specific object, and what does a fast tracker, motion prediction and identity verification add?*

2. Related work

YOLO-World [1] adds a text-conditioned head to a YOLO detector and targets real-time inference. OWL-ViT [2] and OWLv2 [3] use vision transformers with CLIP-style text embeddings [4] and are more accurate but heavier. Grounding DINO [5] is strong on phrase-level queries, at a high computational cost. These works are evaluated on still benchmark images and report accuracy, not the closed-loop behaviour of a robot that has to act on delayed detections. For the fast part of the pipeline we rely on classical components: a correlation-filter tracker [6] and a Kalman filter [7]. We do not claim a new detector or tracker. The contribution is in how these components are combined under delay, in the identity verification, and in the measurement of the whole system on real hardware.

3. Data

There is no large training set, because the detectors are used zero-shot. We record our own video sequences with the UGV camera: about 30 sequences of 10–30 seconds each, in indoor and outdoor scenes. The sequences cover six situations: (S1) slow target motion, (S2) fast motion and abrupt change of direction, (S3) partial occlusion, (S4) the target leaves the frame and returns, (S5) a distractor of the same class is present, (S6) low light. Bounding boxes of the target are annotated in CVAT on every fifth frame and interpolated in between, and a subset is double-checked by a second team member. Sequences are split by scene into validation (thresholds, filter parameters) and test, so that near-identical frames do not leak between them. The camera is calibrated (intrinsic parameters from a checkerboard) so that pixel errors can be converted to angles for the gimbal.

4. Approach and methods

The system has a fast loop, which runs at the camera rate, and a slow loop, which runs the detector at its own rate, possibly on another machine.

What we build ourselves:

- **Asynchronous pipeline.** Each frame is timestamped at capture. The detector receives frames in the slow loop, and its result arrives with a delay. The tracker and the Kalman filter keep a short history of frames and states, so a late detection is applied at the time it refers to and then replayed to the present moment.
- **Motion model with delay compensation.** A Kalman filter (constant-velocity model on the box center and size) fuses tracker and detector measurements, uses the pan and tilt angles of the gimbal to separate

target motion from camera motion, and predicts the target position to compensate for the detection-to-command delay.

- **Gimbal controller.** A controller that converts the predicted pixel error into pan and tilt commands for the ESP32 (through the camera calibration), compared with a plain PID controller that uses only the current error.
- **Identity-aware re-acquisition.** At the first lock we store a visual template of the chosen object, in addition to the text prompt. When the target is lost (low tracker confidence or high filter uncertainty), the gimbal scans around the last predicted position. Each detection candidate is scored by CLIP [4] similarity of its crop to both the text and the stored template, and is accepted only above a threshold tuned on the validation sequences. This distinguishes the target from distractors of the same class.
- **Evaluation harness.** Tools that timestamp every stage, replay recorded sequences with a throttled detector rate, and compute all metrics automatically.

Pretrained components we use as they are: YOLO-World as the main detector, OWLv2 (or OWL-ViT) as a heavier alternative, and optionally Grounding DINO offline, the CLIP model, and an off-the-shelf tracker. The choice of detector determines the detection rate and delay, which are variables of the study.

Compute platforms. The main experiments use a laptop that receives the video stream over Wi-Fi and returns commands. The on-board Raspberry Pi is used to measure the same pipeline at the edge. If an NVIDIA Jetson module (for example, Orin Nano) can be obtained through the UGV club, the detector will run on the robot itself with TensorRT (FP16), and this platform is added to the comparison. The Jetson is an extension and does not block the project.

5. Evaluation

System variants compared on the same scenarios: (V1) gimbal driven directly by the latest detection, without a tracker (baseline); (V2) detector plus tracker and PID; (V3) V2 plus Kalman prediction with delay compensation; (V4) V3 plus CLIP identity verification in re-acquisition (full system). Removing single components from V4 gives an ablation.

Closed-loop metrics (on the robot, at least five runs per scenario and variant): mean and 95th percentile of the centering error in pixels (also in degrees); fraction of time the target is in the frame and near its center; time to first lock after the prompt; time to recover after a loss; rate of locking on a wrong object (S5); end-to-end latency from frame capture to gimbal command. Ground truth is obtained by annotating the frames recorded during the runs.

Open-loop study (replay of recorded sequences): tracking quality (center error and IoU with the annotated box) as a function of the detector rate, which is throttled to several values (for example, 1, 2, 5 and 10 frames per second), for each detector and platform. Detector speed and memory use are reported for the laptop, the Raspberry Pi and, if available, the Jetson.

Goal. With the full system, the target should stay in the frame during at least 90% of the sequence time in scenarios S1–S4 and should be recovered within a few seconds after being lost. We aim to show a clear improvement over the baseline V1. The reported numbers are the result, even if the goal is not met, and failure cases are analysed by scenario.

6. Plan and risks

Before midterm: camera streaming and control of the Pan-Tilt module from Python; camera calibration; latency measurement of the whole chain; baseline V1 working on the robot; recording of the sequences started; availability of a Jetson module confirmed or ruled out. **Before the final:** tracker, Kalman filter and controller (V2, V3); re-acquisition with CLIP (V4); annotation; closed-loop runs and the detector-rate study; ablation and failure analysis; demo. **Risks:** (1) the gimbal does not report its angles or has a large command delay: plan B: use the commanded angles and measure the delay experimentally; (2) the tracker drifts to

the background: plan B: reset it from each new detection and use a second tracker for comparison; (3) CLIP does not separate similar instances: plan B: add a colour and appearance descriptor of the template, and report the limitation; (4) a Jetson module is not available: plan B: continue with the laptop and Raspberry Pi only; (5) annotation takes too long: plan B: reduce the number of sequences and annotate every tenth frame.

7. Contribution statement

- **Arsen Botsko** — gimbal control (ESP32/UART, camera calibration, controller), tracker and Kalman filter with delay compensation, latency measurement, integration on the robot
- **Ivan Zarytskyi** — detector experiments, prompt study, CLIP identity verification and re-acquisition logic
- **Zakhar Veresniuk** — recording and annotation of the sequences, evaluation harness and metrics, documentation, deployment on the Raspberry Pi and Jetson

References

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